Unit: Unit 6: Systems of Linear Equations and Inequalities
Topic: Systems of Linear Inequalities by Graphing

Name: ________________
Date: ____________

Advanced (Habanero)

Q1. What is the graph of the system $x + y \geq 2$ and $x - y \leq 1$?
(A) $x \geq 1, y \geq 1$
(B) $x \leq 1, y \leq 1$
(C) $x \geq 1, y \leq 1$
(D) The entire coordinate plane
(E) Only the origin

Q2. Which of the following is the solution to the system $2x+y \geq 4$ and $x - 2y \leq -3$?
(A) $x \geq 1, y \geq 2$
(B) $x \leq 1, y \leq 2$
(C) $x \geq 1, y \leq 2$
(D) The entire coordinate plane
(E) Only the origin

Q3. The feasible region for the system $x+y \leq 3$ and $x-y \geq -1$ is a
(A) Triangle
(B) Square
(C) Circle
(D) Parallelogram
(E) Rectangle

Q4. What is the corner point of the feasible region for the system $x + 2y \geq 4$ and $2x + y \leq 5$?
(A) $(1, 2)$
(B) $(2, 1)$
(C) $(3, 1)$
(D) $(1, 1)$
(E) $(2, 2)$

Q5. The system $x + y \geq 2$ and $y \geq x + 1$ has
(A) One solution
(B) Infinitely many solutions
(C) No solution
(D) Two solutions
(E) Three solutions

Q6. The system $x - 2y \leq -1$ and $2x + y \geq 4$ has a feasible region that is a
(A) Triangle
(B) Square
(C) Circle
(D) Parallelogram
(E) Rectangle

Q7. Which of the following systems has the same feasible region as the system $x + y \geq 3$ and $x - y \leq 1$?
(A) $x + y \geq 2$ and $x - y \leq 1$
(B) $x + y \geq 3$ and $x - y \leq 2$
(C) $x + y \geq 4$ and $x - y \leq 1$
(D) $x + y \geq 3$ and $x - y \leq 1$
(E) $x + y \geq 2$ and $x - y \leq 2$

Q8. The system $2x + y \geq 5$ and $x - 2y \leq -2$ has a feasible region with
(A) One corner point
(B) Two corner points
(C) Three corner points
(D) Four corner points
(E) Five corner points

Open Ended Questions (FRQ)

Q9. Graph the system $x + y \geq 2$ and $x - y \leq 1$. Identify the corner points of the feasible region.

Q10. Solve the system $2x + y \geq 4$ and $x - 2y \leq -3$ graphically. Write the solution in interval notation.

Chef's Tips

- Tip 1: Ensure students understand the concept of feasible regions
- Tip 2: Encourage students to check their work by plugging in test points
- Tip 3: Remind students to label all axes and provide a title for their graph